

Chapter 5

Premium Calculations

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Most life insurance firms charge gross premiums which are determined by the age of the insured or policyholder attained.

Meanwhile, Net Premiums = Total PV of the policy benefits under the following assumptions:

1. The benefits under the policy will be paid at the ends of the years.
2. The death among the policyholders will occur at exactly the

Let P_x denote the annual premium for an ordinary life policy at age x . Since the PV of this premium is, by definition, equal to the NSP for the benefit provided by the policy

$$P_x \cdot \ddot{a}_x = A_x \quad \Rightarrow \quad P_x = \frac{A_x}{\ddot{a}_x} \quad (5.1)$$

By dividing the policy value with the life annuity, we can calculate the annual payment. This method is known as amortization.

Premiums are the amounts paid to the insurer in exchange for the benefits provided to the insured. The premium may consist of one or more payments.

In traditional life insurance contracts, the premiums would often be a level series of payments made while the policy is in force. Some types of insurance policies (e.g., Universal Life) allow the policyholder a great deal of flexibility with respect to the timing and amounts of the policy payments. Payout annuities are often funded by a single premium.

Net premiums are premiums that, on average, exactly cover the benefit paid by the insurer.

Gross premiums are the premiums actually paid by the insured and are intended to cover expenses and profit, in addition to the benefit.

There are different approaches that can be used in order to determine premiums for a particular benefit. We'll discuss two particular ones here:

- The **Equivalence Principle** states that the EPV (or APV) of the "benefits" must be equal to the EPV of the premiums.
- The **Portfolio Percentile Principle** sets the premium at a level that results in a fixed, specified probability of loss for the insurer, on a block of business, i.e., some large number of such policies.

5.1 Semi-continuous Premium

Semi-continuous insurance is an insurance where the premiums are paid at discrete time points and the death benefits are paid at the moment of death.

Assuming UDD, the semi-continuous premium provided the policy is

$$P(\bar{A}_x) = \frac{\bar{A}_x}{\ddot{a}_x} = \frac{i}{\delta} \frac{A_x}{\ddot{a}_x} = \frac{i}{\delta} P_x \quad (5.2)$$

This equation states that: In general for any life x , the premium $P(\bar{A}_x) \geq P_x$.

Example 5.1.1. You are given:

1. $A_{x:\overline{10}|} = 0.62478$
2. $i = 0.06$
3. ${}_{10}E_x = 0.60269$
4. $\ddot{a}_x = 10.583$
5. Deaths are uniformly distributed over each year of age.

Calculate $P(\bar{A}_{x:\overline{10}|}^1)$.

Solution The term premium is

$$\begin{aligned} P_{x:\overline{10}|}^1 &= \frac{A_{x:\overline{10}|}^1}{\ddot{a}_x} = \frac{A_{x:\overline{10}|} - A_{x:\overline{10}|}^{\frac{1}{2}}}{\ddot{a}_x} \\ &= \frac{0.62478 - 0.60269}{10.583} \\ &= 0.0020873 \end{aligned}$$

$$\begin{aligned} P(\bar{A}_{x:\overline{10}|}^1) &= \frac{i}{\delta} P_{x:\overline{10}|}^1 \\ &= \frac{0.06}{\ln 1.06} \times 0.0020873 \\ &= 0.0021493 \end{aligned}$$

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5.2 Fully Continuous Premium

Theorem 5.1

The continuous premium which paid over h periods is

$${}_h\bar{P}(\bar{A}_x) = \frac{\bar{A}_x}{\bar{a}_{x:\overline{h}|}} \quad (5.3)$$

For pure endowments and term policy, the calculations are similar: divide the policy value with h -payments life annuity.

$${}_h\bar{P}(\bar{A}_{x:\overline{n}|}^1) = \frac{\bar{A}_{x:\overline{n}|}^1}{\bar{a}_{x:\overline{h}|}} \quad (5.4)$$

$${}_h\bar{P}(\bar{A}_{x:\overline{n}|}) = \frac{\bar{A}_{x:\overline{n}|}}{\bar{a}_{x:\overline{n}|}} \quad (5.5)$$

Example 5.2.1. If force of mortality follows deMoivre's law with $\omega = 60$ and $\delta = 0.05$. Find ${}_5\bar{P}(\bar{A}_{20:\overline{10}|})$.

Solution

$$\begin{aligned} \bar{A}_{20:\overline{k}|} &= \bar{A}_{20:\overline{k}|}^1 + {}_kE_{20} \\ &= \int_0^k \nu^t {}_t p_x \mu_{x+t} dt + \nu^k {}_k p_x = \int_0^k e^{-0.05t} \frac{1}{40} dt + \nu^k {}_k p_{20} \\ &= \frac{1}{40} \left[\frac{1 - e^{-0.05k}}{0.05} \right] + e^{-0.05k} \cdot \frac{60 - 20 - k}{40} \\ &= \frac{1}{2} (1 - e^{-0.05k}) + \frac{40 - k}{40} e^{-0.05k} \end{aligned}$$

Hence,

$$\bar{A}_{20:\overline{5}|} = \frac{1}{2} (1 - e^{-0.05(5)}) + \frac{35}{40} e^{-0.05(5)} = 0.79205$$

$$\bar{A}_{20:\overline{10}|} = \frac{1}{2} (1 - e^{-0.05(10)}) + \frac{30}{40} e^{-0.05(10)} = 0.65163$$

$$\begin{aligned} \bar{a}_{20:\overline{5}|} &= \frac{1 - \bar{A}_{20:\overline{5}|}}{\delta} = \frac{1 - 0.79205}{0.05} \\ &= 4.159 \end{aligned}$$

Therefore,

$${}_5\bar{P}(\bar{A}_{20:\overline{10}|}) = \frac{\bar{A}_{20:\overline{10}|}}{\bar{a}_{20:\overline{5}|}} = 0.156679$$

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Example 5.2.2. Using the Standard Ultimate Life Table, find the following:

1. $P_{70:\overline{5}|}^1$
2. $P_{70:\overline{3}|}^1$
3. ${}_5P_{70:\overline{7}|}^1$

5.3 Future Loss

Definition 5.1

The PV of an insurer's future loss is the PV of future outflows from the insurer minus the PV of the future inflows to the insurer.

5.3.1 n -year fully continuous whole life insurance on (x)

For instance, consider a fully continuous whole life insurance with PV of benefit $= S\nu^{T_x}$, and the PV of premium is $\pi\bar{a}_{T_x}$. For this case we can see that the loss at issue is

$${}_0L = \text{PV}(\text{benefit}) - \text{PV}(\text{premium}) = S\nu^{T_x} - \pi\bar{a}_{T_x} \quad (5.6)$$

we again simplify the following

$$\begin{aligned} {}_0L &= \nu^{T_x} - \pi\bar{a}_{T_x} \\ &= \nu^{T_x} - \left(\frac{1 - \nu^{T_x}}{\delta} \right) \\ &= \left(1 + \frac{\pi}{\delta} \right) \nu^{T_x} - \left(\frac{1 - \nu^{T_x}}{\delta} \right) \end{aligned}$$

Taking expectation to the result, this yield to

$$\begin{aligned} \mathbb{E}[{}_0L] &= \bar{A}_x - \pi \left(\frac{1 - \bar{A}_x}{\delta} \right) \\ &= \bar{A}_x - \pi\bar{a}_x \end{aligned}$$

Theorem 5.2 Expected loss of fully continuous whole life premium

$$\mathbb{E}[{}_0L] = \bar{A}_x - \bar{P}(\bar{A}_x)\bar{a}_x \quad (5.7)$$

Theorem 5.3 Variance on loss of fully continuous whole life premium

The variance on issue at loss of fully continuous whole life premium can be compute using the following formulae:

1.

$$V[{}_0L] = \left(1 + \frac{\bar{P}(\bar{A}_x)}{\delta} \right) (\bar{A}_x - \bar{A}_x^2) \quad (5.8)$$

2.

$$V[{}_0L] = \frac{\bar{A}_x - \bar{A}_x^2}{(\delta\bar{a}_x)^2} \quad (5.9)$$

Remark. For **DISCRETE** whole life benefit, replace $\bar{P}(\bar{A}_x)$ with $P(A_x)$, and the force of discount δ will be replace by d .

Example 5.3.1. For whole life annuities-due of 15 per month on each of 200 lives age 62 with independent future lifetimes, you are given:

1. $i = 0.06$

2. $A_{62}^{(12)} = 0.4075$ and ${}^2A_{62}^{(12)} = 0.2105$.

3. π is the single premium to be paid by each of the 200 lives.

4. S is the present value random variable at time 0 of total payments made to the 200 lives.

Using the normal approximation, calculate π such that $\Pr(200\pi > S) = 0.90$.

5.3.2 n -year endowment insurance on (x)

Theorem 5.4 Expected loss of n -year endowment premium

$$\mathbb{E}[{}_0L] = \bar{A}_{x:\overline{n}|} - \bar{P}(\bar{A}_{x:\overline{n}|})\bar{a}_{x:\overline{n}|} \quad (5.10)$$

Theorem 5.5 Variance on loss of n -year endowment premium

The variance on issue at loss of fully continuous whole life premium can be compute using the following formulae:

1.

$$V[{}_0L] = \left(1 + \frac{\bar{P}(\bar{A}_{x:\overline{n}|})}{\delta}\right) ({}^2\bar{A}_{x:\overline{n}|} - \bar{A}_{x:\overline{n}|}^2) \quad (5.11)$$

2.

$$V[{}_0L] = \frac{{}^2\bar{A}_{x:\overline{n}|} - (\bar{A}_{x:\overline{n}|})^2}{(\delta\bar{a}_{x:\overline{n}|})^2} \quad (5.12)$$

5.3.3 n -year term insurance on (x)

For an n -year term insurance which pay 1 at death occurred before the n -th year. The loss at issue is

$${}_0L = \begin{cases} \nu^T - \pi\bar{a}_{\overline{T}|} & 0 \leq T \leq n \\ -\pi\bar{a}_{\overline{n}|} & T > n \end{cases} \quad (5.13)$$

with the term premium

$$\bar{P}(\bar{A}_{x:\overline{n}|}^1) = \frac{\bar{A}_{x:\overline{n}|}^1}{\bar{a}_{x:\overline{n}|}} \quad (5.14)$$

Theorem 5.6 Expected loss of n -year term premium

$$\mathbb{E}[{}_0L] = \bar{A}_{x:\overline{n}|}^1 - \bar{P}(\bar{A}_{x:\overline{n}|}^1)\bar{a}_{x:\overline{n}|} \quad (5.15)$$

Theorem 5.7 Variance on loss of n -year term premium

The variance on issue at loss of n -year term premium can be compute using the following formulae:

1.

$$V[{}_0L] = \left(1 + \frac{\bar{P}(\bar{A}_{x:\overline{n}|}^1)}{\delta}\right) ({}^2\bar{A}_{x:\overline{n}|}^1 - \bar{A}_{x:\overline{n}|}^1{}^2) \quad (5.16)$$

2.

$$V[{}_0L] = \frac{{}^2\bar{A}_{x:\overline{n}|}^1 - (\bar{A}_{x:\overline{n}|}^1)^2}{(\delta\bar{a}_{x:\overline{n}|})^2} \quad (5.17)$$

Example 5.3.2. Calculate the each of the following for life (30) with the amount of benefit 1000, if mortality follows de Moivre's with $\omega = 60$ and $\delta = 0.05$.

1. Ten year term annual premium
2. Ten-payment twenty year term premium.
3. 95% confidence interval for the loss at issue.

5.4 Gross Premium

Some of the typical expenses incurred by life insurance companies include (these are just some examples, not an all-inclusive list):

Expenses incurred at or around the time of issue

- underwriting
- sales / commission
- marketing
- issue

Ongoing Expenses

- overhead / home office
- commission
- premium tax
- maintenance / billing

Claims / termination expenses

- claim payment expenses

In practice, determining and allocating expenses is a time-intensive and non-trivial task, but for our purposes, we'll assume that all expenses are known and given.

Theorem 5.8 Equivalence Principle for Gross Premium

If we wish to use the equivalence principle to determine gross premiums, then we should set gross premiums so that the APV of the benefit amounts paid by the insurer plus the APV of expenses is equal to the APV of the gross premiums paid to the insurer, that is,

$$\text{APV of gross premium} = \text{APV of future benefit} + \text{APV of expenses} \quad (5.18)$$

so that with the equivalence principle, $\mathbb{E}_0[L^g] = 0$.

Example 5.4.1. A person age (85) purchases a three year term policy, with gross premiums payable annually in advance and the death benefit payable at the end of the year of death. The death benefit is \$100,000 plus the sum of the gross premiums paid. You are given:

$$q_{85} = 0.2 \quad q_{86} = 0.3 \quad q_{87} = 0.4 \quad i = 10\%$$

Expenses consist of:

- 30% of the initial gross premium and 2% of renewal gross premiums.
- \$500 at issue

Using the equivalence principle, calculate the gross annual premium for this policy.

Example 5.4.2. For a fully discrete whole life insurance of 1000 on (x) , you are given:

- The following expenses are incurred at the beginning of each year:
- An additional expense of 20 is paid when the death benefit is paid

	Year 1	Year 2+
Percent of premium	75%	10%
Maintenance expenses	10	2

- The gross premium is determined using the equivalence principle
- $i = 0.06$
- $\ddot{a}_x = 12.0$
- ${}^2A_x = 0.14$

Calculate the variance of the loss at issue random variable to the nearest 100. [Ans: 88,900]

Example 5.4.3. You are given the following information about a whole life policy purchased from a life

Gross Premium: P paid at the beginning of each year that (x) is alive.

Death Benefit: \$50,000 plus the sum of all gross premiums paid by the insured over the course of the policy, all payable at the end of the death.

Premium Tax Expenses: 2% of gross premiums paid (incurred at the time of premium payment)

Issue Expenses: \$5,000 at the time of policy issue.

Maintenance Expenses: \$250 at the beginning of each year (including the first year)

Termination Expenses: \$1,000 at the time of death benefit payment

The annual gross premium P is determined using the Equivalence Principle. You are given:

$$i = 6\%, \quad A_x = 0.16079, \quad (IA)_x = 4.17226$$

You are given the following additional information:

$$A_{x+10} = 0.24338, \quad (IA)_{x+10} = 5.8923, \quad \ddot{a}_{x+10} = 13.367$$

1. Calculate A_x .
2. Calculate P .
3. Calculate the net premium reserve at time 10.

5.5 Profits

Profits can be incorporated either implicitly or explicitly into the calculation of life insurance premiums. Writing out expressions for our future loss random variables, we can see that their values depend on when the insured dies. We can examine the future loss as a function of the future lifetime of the policyholder.

In general, for a life insurance policy, the insurer's loss will tend to decrease with the future lifetime of the insured. For annuities, the reverse will generally be true.

5.6 Portfolio Percentile Premium Principle

The Portfolio Percentile Premium Principle gives us a way to set premiums so that the probability of losing money on the entire block of business is set to a fixed value. Assume we have N policies with independent and identically distributed (iid) future loss random variables ${}_tL_1, \dots, {}_tL_N$. Then we define the total future loss of the block as

$$L = \sum_{i=1}^N {}_tL_i \quad (5.19)$$

By the Central Limit Theorem, for large N , L has an approximately Normal distribution:

$$L \sim^{\text{approx}} \mathcal{N}(N \times \mathbb{E}[L], N \times V[L]) \quad (5.20)$$

we can also scale and shift this into a standard normal distribution:

$$\frac{L - (N \times \mathbb{E}[L])}{\sqrt{N \times V[L]}} \sim \mathcal{N}(0, 1) \quad (5.21)$$

This allows us to more easily calculate the desired probabilities.

Example 5.6.1. An insurer issues 100 fully discrete whole life policies to independent persons age (x). Assume that

$$d = 0.06 \quad A_x = 0.4 \quad {}^2A_x = 0.2$$

The policies are distributed as follows:

Face amount	Number of Policies	Annual Premium per policy
100,000	80	5,000
400,000	20	19,000

Using a normal approximation, find the approximate probability that the present value of the insurer's profits exceeds 4,000,000.